

The Security Premium: Why Taiwan Maintains the World's Most Restrictive Policy toward Chinese Electric Vehicles¹

DPI-410: The Politics of Development | Harvard Kennedy School

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Spring 2026 — Final Paper

Abstract. Taiwan produces only 275,000 vehicles a year, hosts no domestically owned EV brand, and has no organized producer lobby. The Grossman–Helpman model predicts minimal restriction. Its observed policy is a complete ban on finished Chinese vehicles — the most restrictive of any economy in the world. We argue the missing variable is perceived national security: when the import source is a geopolitical adversary capable of weaponizing economic interdependence, governments apply a security premium above the Grossman–Helpman equilibrium. Three methods triangulate on this conclusion. A cross-national ordered logit shows Taiwan is the largest outlier in the industry-only model; once the EU is treated as a single decision-maker, the security channel achieves statistical significance. A synthetic control shows Taiwan's production fell 29 percent below the counterfactual after 2016. A within-case process trace closes the argument: tightening began only when the DPP raised security salience — the same structural threat under the KMT produced no restriction. Security salience, not security threat, moves policy.

Contents

1. Introduction

2. Theoretical Framework

- 2.1 The Industry Channel: Protection as a Function of Lobbying
- 2.2 The Security Channel: Perceived Threat, Historical Incidents, and the Commitment Problem

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- 2.2a Direct Incidents of Economic Coercion
- 2.2b Military Escalation
- 2.2c Domestic Political Reinterpretation
- 2.3 Consumer Welfare as a Scope Condition: The Q4 Boundary
- 2.4 Synthesis: The 2x2 Framework and Interaction Logic

3. Data and Research Design

4. Cross-National Results

5. Taiwan: A Within-Case Process Trace

6. Adjacent Comparisons and Future Research

7. Conclusion

References

Appendix A: Data Sources and Variable Construction

Appendix B: Robustness Checks

Appendix C: Replication Script

1. Introduction

In 2023, China surpassed Japan as the world’s largest automobile exporter. By 2025, BYD had overtaken Tesla in global battery-electric vehicle sales, delivering 2.25 million units. China’s new energy vehicle penetration crossed 50 percent of new car sales by mid-2024, three years ahead of the State Council’s 2035 target. The Kiel Institute estimates BYD alone received \$2.1 billion in state subsidies in 2022, roughly 3.5 percent of firm revenues (Kiel Institute for the World Economy 2024).

The policy responses to this export surge have been remarkably divergent. The United States imposed cumulative tariffs exceeding 150 percent (USTR 2024). The European Union imposed countervailing duties of 17–35.3 percent. Yet Japan applies a zero tariff. Australia applies zero under the China-Australia Free Trade Agreement. Brazil imposes 18 percent while welcoming BYD’s \$1 billion factory in Bahia. This variation is the puzzle that motivates this paper.

The canonical model of endogenous trade protection — Grossman and Helpman (1994) — predicts higher tariffs where domestic producers are politically organized and face rising import penetration. This explains the United States and the EU, where large incumbent automakers lobbied for protection. But it cannot explain Japan, which has the second-largest auto sector in the world and zero tariffs on Chinese EVs. Nor can it explain the most striking anomaly: Taiwan, which produces only 275,000 vehicles annually and hosts no domestically owned EV brand, yet maintains the most restrictive policy of any economy: a complete ban on finished Chinese vehicles.

This paper argues that the missing variable is national security. When the import source is a geopolitical adversary capable of weaponizing economic interdependence, governments internalize expected future coercion costs into the trade-policy objective function, generating what we call a security premium above the Grossman-Helpman equilibrium. Taiwan is the crucial case for this argument: it scores low on every industry-pressure variable that the standard model associates with protection, yet scores maximally on security threat from the import source.

Research Question: Why does Taiwan — which has one of the smallest domestic auto industries among major economies — maintain the most restrictive policy toward Chinese EV imports in the world?

Hypotheses:

- **H1 (Industry baseline):** Countries with larger domestic auto sectors adopt more restrictive Chinese EV policies, controlling for income and trade dependence.
- **H2 (Security premium):** Countries facing higher security threat from China adopt more restrictive policies, after controlling for industry size. Taiwan’s maximum restrictiveness

despite minimal industry pressure constitutes the strongest evidence for this channel.

The paper combines a cross-national ordered logit analysis of 48 automobile-producing countries with a within-case process trace of Taiwan. The cross-national regression establishes that Taiwan is the largest positive residual in the industry-only model, and that the security channel achieves statistical significance in specifications that correctly account for EU clustering — where 15 member states share a single Commission policy rather than making independent decisions. The process trace provides mechanism identification: security escalations precede policy tightening, official rationale cites national security rather than industrial policy, and the Ma Ying-jeou counterfactual shows that structural security threat alone is insufficient without political activation.

2. Theoretical Framework

2.1 The Industry Channel: Protection as a Function of Lobbying

The Grossman–Helpman (1994) model of endogenous trade policy provides the baseline prediction. Organized interest groups bid for tariff protection through policy-contingent campaign contributions, and politicians maximize a weighted sum of contributions and aggregate welfare. In equilibrium, sectors that are politically organized, face high import penetration, and have low import-demand elasticity receive the most protection (Grossman and Helpman 1994). Applied to the Chinese EV case, the model predicts that countries where domestic auto manufacturers are large, employ substantial labor, and face rising Chinese import penetration should adopt the most restrictive policies. The United States (10.6 million vehicles), the EU (12.8 million), and India (6.0 million) fit this prediction.

But the model fails on two fronts. First, it overpredicts protection for Japan and South Korea. Japan produces 8.2 million vehicles annually and applies a zero tariff on Chinese EVs. These globally competitive automakers do not perceive Chinese EVs as an existential threat — firms that

export at scale into the Chinese market face retaliatory market closure if their home governments restrict Chinese imports, suppressing lobbying even in large sectors. Second, and more critically, the model underpredicts protection for Taiwan — the anomaly identified in Section 1 — where no domestically-owned national champion exists to serve as the GH lobbying actor.

2.2 The Security Channel: Perceived Threat, Historical Incidents, and the Commitment Problem

Our explanation pulls together two theoretical concepts. The first is Farrell and Newman's (2019) idea of weaponized interdependence: countries at the center of global economic networks can use their position politically, spying on flows running through their network (the panopticon effect) and cutting those flows off (the chokepoint effect). The second is the commitment problem (Blattman 2022; Fearon 1995): when the distribution of power between parties is shifting over time, today's agreement cannot bind tomorrow's actor, so the side losing leverage has an incentive to act preemptively.

Modern battery-electric vehicles intensify both vulnerabilities relative to prior automotive trade disputes. They are software-defined platforms with over-the-air update capabilities, GPS tracking, and LiDAR-based environmental mapping — creating information-security vulnerabilities with no analogue in previous automotive trade disputes. Simultaneously, the battery supply chains underpinning Chinese EV cost competitiveness are concentrated in China at shares that create material chokepoint exposure for any country that allows deep EV fleet penetration without domestic supply-chain alternatives.

Structural vulnerability alone cannot explain observed policy variation: many countries share broadly similar structural exposure to Chinese supply chains without enacting maximum restriction. The transformation of structural vulnerability into active trade restriction requires a political process through which governments come to perceive the vulnerability as sufficiently threatening to justify the consumer-welfare costs of restriction. This perception-building process operates through three formally identified sources.

2.2a Direct Incidents of Economic Coercion The first source is hard evidence that the exporter is willing to use trade as political punishment. China’s interventions include: tourism to Taiwan collapsed after the DPP’s 2016 win (Chinese visitors fell from 4.2 million in 2015 to about 700,000 by 2019); Australian wine, barley, coal, and lobster faced sudden barriers after Canberra called for a COVID-19 origins investigation in 2020; Lithuanian exports were effectively frozen at Chinese customs in 2021 after Vilnius allowed Taiwan to open a “Taiwanese Representative Office”; South Korean tourism, retail, and entertainment were squeezed after the THAAD anti-missile system was deployed in 2017. Collectively, these episodes made clear that commercially important ties with China are conditional on political acquiescence. For a government deciding whether to deepen dependence on Chinese EVs, that record matters.

2.2b Military Escalation The second source is the broader security backdrop. The Taiwan case is the limiting example: China’s People’s Liberation Army (PLA) encirclement exercises after the August 2022 Pelosi visit, the 2024 “Joint Sword” drills following President Lai’s inauguration, PLA aircraft entering Taiwan’s air defense identification zone almost daily, and the publication of detailed PLA operational concepts for blockade and quarantine all raise the perceived probability that future commercial dependencies will be activated as instruments of pressure. Military escalation does not by itself produce trade restriction — many countries face elevated military risk without restricting commerce — but it converts ambient economic-coercion incidents into politically actionable evidence.

2.2c Domestic Political Reinterpretation The third source is domestic. Whether the perceived threat translates into restrictive policy depends on which domestic actors hold office, how they frame the threat, and how voters respond to that frame. The same structural threat can produce different policy outcomes depending on whether the governing party is electorally invested in heightened threat salience or in normalized cross-strait commerce. This is where Pierson’s (1993) policy-feedback logic and the broader literature on political entrepreneurship enter: actors who

benefit from a securitized framing of cross-strait commerce have incentives to make coercion incidents publicly salient and to translate them into formal policy instruments, while actors invested in normalization have incentives to downplay the same events. The Ma Ying-jeou counterfactual developed in Section 5 is the cleanest empirical illustration of this dynamic: trade dependence on China grew throughout the KMT's 2008–2016 tenure without triggering additional restriction, even though every structural feature of the security situation — territorial claim, military capability, geographic proximity — was identical to the post-2016 environment in which restriction tightened sharply. Security threat is constant; security salience is variable, and it is the variable that matters for policy.

2.3 Consumer Welfare as a Scope Condition: The Q4 Boundary

The security channel predicts that small-industry countries facing a serious security threat will close their markets. But not all small-industry countries observe this behavior. Norway, Denmark, and Vietnam apply minimal barriers on Chinese EVs despite having little to protect — because consumers, who lose from protection, finally get a hearing when there is no organized producer drowning them out.

The Grossman-Helpman framework contains this prediction. Consumer interests are always in the government's objective function, weighted alongside campaign contributions. But because the benefits of protection are concentrated on a few producers while the costs are spread thin across millions of consumers, consumers almost always lose the political fight. The logic is Mancur Olson's (1965) classic collective-action problem: small concentrated groups organize easily; large diffuse groups do not. Take the producer lobby out of the picture, and the asymmetry collapses. The cost of restriction to consumers stays the same, but there is no longer a concentrated political beneficiary to offset it.

This effect is amplified in countries where EVs are already popular. Consumers there have visible preferences for electric vehicles; the price gap between affordable Chinese EVs and the alternatives is salient; and restricting access to that cheaper option imposes a clear, attributable cost. With no

producer lobby pushing back, openness becomes the politically dominant outcome. The implication is a scope condition on H2: the security premium is triggered only when the perceived threat is severe enough to override the consumer-welfare cost of restriction. For most small-industry, high-EV-adoption countries, that threshold is not reached.

2.4 Synthesis: The 2x2 Framework and Interaction Logic

The lobbying channel, the security channel, and the consumer-welfare scope condition fit together into a single two-by-two typology. The two axes are the size of the domestic internal-combustion (ICE) auto industry and how far EV adoption has progressed. Each combination produces a different politics.

Table 1. The 2x2 Framework: ICE Industry Size × EV Adoption Level. Thresholds: ICE production $\geq 1,000\text{K}$ vehicles (OICA 2022); EV $\geq 5\%$ of new car sales (IEA average 2022–24). Quadrant assignments are mechanical, based on these thresholds.

	Low EV Adoption ($<5\%$)	High EV Adoption ($\geq 5\%$)
Large ICE ($\geq 1\text{M}$)	Q1: Latent Threat — Japan, India, Mexico, Brazil. GH predicts preemptive lobbying; actual restriction varies widely.	Q2: Active Threat — USA, Germany, South Korea, France. GH operative: incumbents threatened and mobilized.
Small ICE ($<1\text{M}$)	Q3: No Industry Channel — Taiwan, Pakistan, South Africa, Russia. No producer lobby; security channel necessary and sufficient.	Q4: Consumer Surplus — Norway, UK, Vietnam, Denmark. Consumers want cheap EVs; high political cost of restriction.

The key empirical test becomes whether the industry-restrictiveness slope differs across quadrants. The econometric implementation is an interaction term: the marginal effect of security on restrictiveness should be largest in Q3, where the industry channel is absent. Taiwan sits in Q3 — small ICE industry, low EV adoption — with maximum restrictiveness ($DV=3$), constituting the cleanest test case for H2.

The regression specification derived from this framework is:

$$Restrictiveness_i = f(\beta_1 \cdot Industry_i + \beta_2 \cdot Security_i + \beta_3 \cdot Industry_i \times Security_i + \gamma X_i + \epsilon_i)$$

Four testable predictions follow. OI-1: Across the full sample, bigger industries should be associated with more restriction, but the relationship should be much stronger in Q2 than in Q3. OI-2: Within Q3, restrictiveness should vary with security alignment rather than production; Taiwan, the only high-restriction country in this quadrant, is the outlier. OI-3: Taiwan should be the largest outlier in the industry-only model. OI-4: Q4 countries should cluster at low restriction.

3. Data and Research Design

Sample and Dependent Variable

We code policy restrictiveness toward Chinese EV imports for 193 sovereign states on an ordinal 0–3 scale. The analytical sample for regression comprises 48 automobile-producing countries (OICA members with non-zero production). Coding follows the logic of Kee et al. (2009): 0 = Open/MFN at or below 10 percent; 1 = Soft protection (mild tariffs 10–25 percent, subsidies); 2 = Tariff escalation (above 25 percent or CVD); 3 = NTB or de facto ban (above 100 percent effective or administrative prohibition). Sources include the Global Trade Alert database, WTO Trade Policy Reviews, and national regulatory announcements.

As a secondary specification, we construct a continuous ad valorem equivalent (AVE) that assigns each country its effective tariff rate, with Taiwan coded at 200 percent (a conservative estimate of the prohibitive-tariff equivalent of a complete ban). OLS with heteroskedasticity-robust standard errors replaces ordered logit for this specification.

Table A.1 (Appendix A) shows the DV distribution spread across all four categories. The small number of DV=3 observations (N=7) includes the United States, India, and Taiwan — making Taiwan’s position particularly anomalous given its industry size. Table A.2 (Appendix A) reports summary statistics for the auto-producing analytical sample; Taiwan’s position within the DV=3 category is particularly revealing: at 275K vehicles, it sits an order of magnitude below the category

mean of 4,210K. Mean alliance ordinal scores increase monotonically with restrictiveness from DV=0 to DV=3, providing preliminary descriptive support for H2.

Independent Variables

Industry channel (A.1): Log of domestic motor vehicle production (OICA, 2022 — pre-treatment for most countries, as the universal shock year is 2023 when BYD’s global exports surged).

Security channel (C.1): Alliance ordinal score (0–5), measuring the strength of a country’s security relationship with the United States relative to China. The scale runs from 0 (China-aligned SCO member) through 2 (limited partner), 3 (security cooperation), 4 (major non-NATO ally equivalent), to 5 (formal treaty ally). Taiwan is coded 4 (Major Non-NATO Ally equivalent) reflecting the Taiwan Relations Act, arms sales, and de facto alliance relationship. This is our primary security measure because it is available for all countries including Taiwan, which lacks a UNGA score due to UN non-membership (Bailey et al. 2017).

Controls: V-Dem Liberal Democracy Index (D.1) (V-Dem Institute 2025), log GDP per capita (World Bank WDI), BRI membership (B.3), and a global competitive dummy (Japan, South Korea, Germany = 1).

Taiwan’s UNGA Exclusion

Taiwan is not a UN member and therefore has no Bailey-Strezhnev-Voeten ideal point score. Our solution: the primary specification (Option A) uses alliance ordinal only, which keeps Taiwan in the sample. As robustness, we estimate with UNGA included (Option B, Taiwan excluded), then predict Taiwan’s DV out-of-sample using estimated coefficients and an imputed UNGA value.

Regression Specification

We estimate four ordered logit models of increasing complexity: M1 (Industry only), M2 (Security only), M3 (Combined), M4 (Full controls), M5 (Interaction), M6 (+ Global competitive dummy), and M7 (Full + Global competitive dummy).

EU Clustering

The EU presents a clustering problem: approximately 15 member states all have $DV=2$, reflecting a single policy decision by the European Commission (Regulation 2024/2754). We present results three ways: (a) EU members as individual observations ($N \approx 48$), (b) EU collapsed to one production-weighted observation ($N \approx 35$), (c) EU excluded entirely ($N \approx 35$).

4. Cross-National Results

4.1 Descriptive Patterns

Figure 1 plots policy restrictiveness against log industry size. Across most of the production range the Grossman-Helpman prediction holds, but the two corners of the scatter are where the theory breaks down: Japan and South Korea sit in the lower right (large industry, low restriction) and Taiwan in the upper left (small industry, maximum restriction).

Figure 2 maps countries into the 2x2 quadrant space. The Q4 cluster (small ICE, high EV adoption) is dominated by $DV=0$ countries, confirming that consumer demand for cheap EVs creates political costs for restriction when no domestic industry lobby is present. Taiwan sits firmly in Q3 (small ICE, low EV adoption) yet is the only $DV=3$ country in that quadrant — the visual representation of the security premium.

Table 3: 2x2 Framework Quadrant Summary

Quadrant	N	Mean DV	Mean Prod (K)	Mean EV %
Q1: Latent Threat	9	1.78	2,905.5	2.17
Q2: Active Threat	7	1.43	3,334.8	8.79
Q3: No Industry Channel	22	1.36	264.2	2.07
Q4: Consumer Surplus	10	1.50	274.5	16.23

Table 4: Cross-Tabulation: 2x2 Quadrant $\times$ Policy Restrictiveness

Quadrant	Open	Soft	Tariff	NTB/Ban
Q1: Latent Threat	1	2	4	2
Q2: Active Threat	2	1	3	1
Q3: No Industry Channel	3	9	9	1
Q4: Consumer Surplus	2	1	7	0

Within Q3, Taiwan's DV=3 is the sole NTB case in the quadrant where the industry channel is absent — the cleanest evidence for the security premium.

4.2 Ordered Logit Results

M1 (Industry only) confirms that log production has a positive coefficient, consistent with Grossman-Helpman. M2 (Security only) shows a near-zero alliance coefficient in the full sample (+0.02, SE=0.18), reflecting the opposing pulls of globally competitive automakers (high alliance, low restriction) and China-aligned high-restriction countries. M3 (Combined) is the critical test: the security coefficient is -0.088 (SE=0.190, LR test M1 vs. M3: $p=0.643$). The sign is negative, but this does not constitute evidence against H2 — among high-alliance countries, Japan, South Korea, Germany, and Australia are all open, outnumbering Taiwan and pulling the regression slope downward. The EU clustering problem compounds this further.

M6 addresses the globally competitive exception by adding the dummy for Japan, South Korea, and Germany. The dummy is large and negative (-3.069 , SE=1.439, $p<0.05$), confirming these countries are systematically more open than their industry size predicts. The LR test comparing M3 to M6 is significant ($p=0.027$), and M7 achieves the best model fit (AIC=111.4, Table 6). But even after absorbing the globally competitive exception, the security coefficient remains insignificant in the full sample. The EU clustering problem is what prevents the signal from emerging.

Table 5: Ordered Logit Results (Option A: Alliance Ordinal Only, Taiwan Included)

Model	Term	Estimate	SE	z value	sig
M1: Industry	log_prod	0.194	0.126	1.53	

Model	Term	Estimate	SE	z value	sig
M2: Security	alliance_ord	0.018	0.179	0.10	
M3: Combined	log_prod	0.215	0.134	1.60	
M3: Combined	alliance_ord	-0.088	0.190	-0.46	
M4: Full	log_prod	0.181	0.145	1.25	
M4: Full	alliance_ord	-0.240	0.257	-0.94	
M4: Full	vdem	0.284	1.958	0.15	
M4: Full	log_gdp	0.345	0.435	0.79	
M5: Interaction	log_prod	0.628	0.352	1.78	*
M5: Interaction	alliance_ord	1.441	1.218	1.18	
M5: Interaction	log_prod:alliance_ord	-0.126	0.099	-1.27	
M6: + Global Comp	log_prod	0.303	0.143	2.13	**
M6: + Global Comp	alliance_ord	-0.052	0.193	-0.27	
M6: + Global Comp	global_comp	-3.069	1.439	-2.13	**
M7: Full + Global Comp	log_prod	0.271	0.154	1.76	*
M7: Full + Global Comp	alliance_ord	-0.228	0.264	-0.87	
M7: Full + Global Comp	vdem	0.270	2.012	0.13	
M7: Full + Global Comp	log_gdp	0.439	0.449	0.98	
M7: Full + Global Comp	global_comp	-3.275	1.447	-2.26	**

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Table 6: Model Fit Comparison (LR test M1 vs M3: $p=0.643$; LR test M3 vs M6: $p=0.027$)

Model	AIC	LogLik	N
M1: Industry	116.7	-54.4	47
M2: Security	119.1	-55.6	47
M3: Combined	118.5	-54.3	47
M4: Full	114.9	-50.4	46
M5: Interaction	118.9	-53.4	47
M6: + Global Comp	115.6	-51.8	47
M7: Full + Global Comp	111.4	-47.7	46

Table 7: Predicted vs. Observed Policy Restrictiveness for Key Cases. Taiwan has the largest positive residual under M1 (+1), unchanged under M3, reflecting that the combined model cannot reduce Taiwan's anomaly through the security coefficient alone — consistent with the aggregate null result and the EU-corrected results below.

Country	ISO	Obs DV	Pred M1	Resid M1	Pred M3	Resid M3
Australia	AUS	0	2	-2	1	-1
Japan	JPN	0	2	-2	2	-2
South Korea	KOR	0	2	-2	2	-2
Thailand	THA	0	2	-2	2	-2
Vietnam	VNM	0	2	-2	2	-2
Brazil	BRA	1	2	-1	2	-1
India	IND	3	2	+1	2	+1
Iran	IRN	3	2	+1	2	+1
Pakistan	PAK	1	2	-1	2	-1
Taiwan	TWN	3	2	+1	2	+1
United Kingdom	GBR	1	2	-1	2	-1
Germany	DEU	2	2	0	2	0
Russia	RUS	2	2	0	2	0

When the EU is collapsed to a single production-weighted observation, the alliance coefficient becomes -0.524 ($SE=0.231$, $t=-2.27$, $p<0.05$) — statistically significant and in the direction predicted by H2 for independently policy-setting countries. EU exclusion produces a near-identical result (-0.552 , $SE=0.235$, $t=-2.35$, $p<0.05$), confirming EU coding is not driving the full-sample null.

Table 8: EU Clustering Robustness: M3 (log_prod + alliance_ord) Across Three EU Coding Strategies

Specification	Term	Estimate	SE	t
Full Sample (N=47)	Log Production	0.215	0.134	1.60
Full Sample (N=47)	Alliance Ordinal	-0.088	0.190	-0.46
EU Collapsed (N=32)	Log Production	0.239	0.140	1.71*
EU Collapsed (N=32)	Alliance Ordinal	-0.524	0.231	-2.27**
EU Excluded (N=31)	Log Production	0.198	0.142	1.40
EU Excluded (N=31)	Alliance Ordinal	-0.552	0.235	-2.35**

*** $p<0.01$, ** $p<0.05$, * $p<0.10$

Figure 3 plots the M2 and M3 log-odds for each variable with 95% confidence intervals, showing

that the alliance coefficient is not displaced when log production enters the specification — the coefficients are directionally stable, confirming the two channels are not collinear. The regression results establish that the full-sample null is a data-structure problem, not an absence of signal; the security channel achieves significance precisely when the EU clustering distortion is removed. The residual plot in the next section makes Taiwan’s anomalous position visible in a way that the regression table cannot.

4.3 The Residual Plot: Visualizing the Security Premium

Figure 4 is the paper’s core empirical visualization. The y-axis is the residual from an OLS regression of the 0–3 DV on log production alone: positive values indicate countries that are more restrictive than their industry size predicts. Taiwan appears in the top-right quadrant — maximum restriction, smallest industry, highest security alignment. Japan, South Korea, and Germany appear in the bottom-right (globally competitive exception). The near-zero aggregate slope ($b = -0.05$, $SE = 0.08$) reflects the cancellation of these two opposing clusters rather than an absence of a security premium. The argument rests on Taiwan’s position, not on the slope of the aggregate trendline.

The leave-Taiwan-out cross-validation (Table 9) confirms Taiwan’s position is not a statistical artifact. Estimated on the remaining 46 countries, the model assigns only a 4.0 percent probability to Taiwan receiving DV=3.

Table 9: Leave-Taiwan-Out Cross-Validation

Observed DV	3 (NTB/Ban)
Predicted Class	2
P(DV=0)	0.165
P(DV=1)	0.289
P(DV=2)	0.507
P(DV=3)	0.040

Estimated without Taiwan; Taiwan’s DV predicted out-of-sample.

The Q3 subsample provides the cleanest cross-national test of H2: within Q3 — countries where

the industry channel is absent by construction — the alliance coefficient is 0.209 (SE = 0.254, $p = 0.41$, $N = 22$). The small subsample limits statistical power, but Taiwan’s position as the sole DV=3 country in Q3 remains analytically informative.

Table 10: Security Channel Test: Q3 Subsample (Small ICE, Low EV, $N = 22$)

Term	Estimate	SE	t	p	sig
Alliance Ordinal	0.209	0.254	0.82	0.41	

4.4 Robustness

Full robustness checks are reported in Appendix B (§B.1–B.4). Taken together, they yield a nuanced picture. The full-sample M3 specification does not confirm H2 as a statistically significant coefficient (LR test $p=0.64$): the null result reflects the opposing pulls of the globally competitive automaker exception (Japan, Korea, Germany: high alliance, low restriction) and the EU clustering problem. When either issue is addressed — EU collapsed or excluded — the security channel achieves significance at the $p<0.05$ level among countries independently setting their own policies. The paper’s empirical argument therefore rests on three triangulated claims: Taiwan’s position as the largest positive residual in Figure 4, the leave-Taiwan-out validation confirming genuine out-of-sample anomaly, and the within-case evidence in Section 5.

5. Taiwan: A Within-Case Process Trace

Taiwan functions as a crucial case in the sense of Eckstein (1975): given the Grossman-Helpman model’s strong prediction of low restriction, Taiwan’s observed maximum restrictiveness constitutes the most demanding test of the security channel. This section traces the causal process through five empirical dimensions, complemented by a synthetic control analysis that quantifies the production-side footprint of security-driven tightening.

5.1 Declining Domestic Capacity

Taiwan's automotive sector has been in structural decline for three decades. Total output peaked at approximately 900,000 units in the early 1990s and has fallen to roughly 275,000 by 2024. The industry is dominated by Japanese-affiliated assemblers — Kuozui (Toyota), China Motor Corporation (Mitsubishi), and Ford Lio Ho — none domestically owned in the meaningful sense. This is precisely the profile the Grossman-Helpman model associates with low protection (Grossman and Helpman 1994, 1995). The absence of a domestically owned national champion eliminates the primary Grossman-Helpman lobbying actor. No Taiwanese firm has the organizational interest or political capacity to lobby for protection of its own Chinese EV market share.

5.2 Rising Cross-Strait Trade Dependence

While domestic production declined, Taiwan's trade dependence on China in the automotive sector grew dramatically. Imports of vehicles and transport equipment (HS 17) from China rose from under \$200 million in 2003 to \$1.75 billion in 2022. The reversal since 2022 is striking: imports declined sharply, coinciding with political tightening and the August 2024 local-content rule. This temporal correspondence — trade dependence growing for two decades, then abruptly reversing when security salience spikes — is consistent with the security channel. The import growth created precisely the vulnerability that the commitment problem predicts: once downstream assemblers incorporated Chinese components, switching costs began accumulating.

5.3 The Parts-Import Substitution Pattern

Japan's share of Taiwan's auto-parts imports fell from 70 percent in 2004 to 24 percent in 2025. China surged from under 5 percent to over 32 percent, surpassing Japan as the largest source. The 35 percent local-content requirement announced in August 2024 was explicitly designed to close the 'MG model' loophole, where China Motor Corporation assembled MG-branded vehicles using predominantly Chinese components. The deeper motivation was strategic: halting growing dependency before it became structurally irreversible at the firm level — a commitment-problem

response, though one triggered reactively once dependency had already reached 32 percent rather than preemptively at the outset. This timing is theoretically consistent: the commitment problem predicts intervention when switching costs become salient, not necessarily before any dependence accumulates. Taiwan’s policymakers appear to have identified 2024 as the threshold at which further integration would make reversal prohibitively costly — consistent with the MG model loophole representing the tipping point where Chinese components crossed from supplementary to structural (Blattman 2022; Farrell and Newman 2019). The regulation targeted not the EV sector per se but the structural condition that would make the security threat material: deep supply-chain integration with a geopolitically adversarial state.

5.4 Beijing’s Demonstrated Willingness to Weaponize Interdependence

The credibility of the security channel depends on whether economic coercion is plausible. After the DPP’s 2016 presidential victory, Beijing restricted group tourism to Taiwan. Chinese visitor arrivals collapsed from 4.2 million in 2015 to approximately 700,000 by 2019. Other-origin visitors recovered fully after COVID-19, confirming that the Chinese decline was a deliberate political restriction rather than a global travel shock. Taiwan’s policymakers observed this in real time: Beijing demonstrated willingness to impose economic costs for political reasons. This is the empirical foundation for the security channel’s theoretical mechanism — the demonstration effect that makes the commitment problem credible rather than abstract.

5.5 Temporal Sequencing: Security Events and Policy Changes

The strongest evidence comes from the temporal sequence: security escalations precede policy tightening, and official rationale cites information security and cross-strait political risk rather than automotive industrial policy. The original ban on finished Chinese vehicles (2002) was an administrative designation under the Mainland Affairs Council’s cross-strait trade framework — a political-security instrument, not a trade-defense measure. The 2024 local-content rule followed PLA encirclement exercises after President Lai’s inauguration. The proposed 2025 third-country

import ban explicitly cited information security (US Commerce Department 2024).

The counterfactual is also informative: during 2010–2015 under the KMT’s Ma Ying-jeou administration, trade dependence grew without triggering additional restrictions. It was only after the DPP’s 2016 victory and the subsequent deterioration in cross-strait relations that the government began actively tightening policy — even as the economic fundamentals (small industry, growing consumer demand) would have predicted liberalization.

The Ma Ying-jeou Counterfactual. Under KMT governance (2008–2016), Taiwan’s automotive trade dependence on China grew without triggering additional restrictions, despite a structurally identical security situation — same PRC territorial claim, same military capability, same cross-strait proximity. Only the political salience of the security threat changed with the DPP’s 2016 victory. This distinction between structural threat (constant) and security salience (variable, party-driven) is the paper’s strongest single piece of evidence against reverse causality.

5.6 Synthetic Control: Quantifying the Security-Channel Effect

The process trace establishes the causal sequence qualitatively. We complement it with a synthetic control analysis (Abadie, Diamond, and Hainmueller 2010) that quantifies the production-side footprint of security-driven policy tightening. The method constructs a counterfactual Taiwan — a weighted average of comparable small-ICE countries that matched Taiwan’s pre-2016 production trajectory — and measures post-2016 divergence as the estimated effect of securitization.

Synthetic Taiwan is primarily composed of Argentina (weight = 0.384) and South Africa (weight = 0.235), with nine other donor countries contributing smaller weights (Table 16). The pre-treatment fit is excellent: RMSPE = 0.042 log units (4.3% in production terms), confirming the counterfactual is credible (Table 17).

Figure 6 shows the path plot: Taiwan’s actual production tracks the synthetic counterfactual closely through 2015, then diverges sharply downward from 2016 onward. Figure 7 quantifies the gap:

post-2016 through 2019 (before COVID confounds inference), Taiwan produced on average 0.256 log units — approximately 29% — below what the counterfactual predicts. Placebo tests (Figure 8) show Taiwan’s gap sits in the lower tail of the donor distribution through 2019.

Table 15: Predictor Balance: Taiwan vs. Synthetic Taiwan. Pre-treatment RMSPE = 0.042 log units (4.3% in production terms).

Predictor	Taiwan	Synthetic Taiwan	Donor Mean
GDP per capita (mean, pre-2016)	10,628.7	10,699.7	10,628.7
Log production (2010)	12.623	12.622	11.761
Log production (2012)	12.734	12.733	11.916
Log production (2014)	12.846	12.818	12.248

Table 16: Synthetic Control Donor Weights.

Country	ISO	Weight
Argentina	ARG	0.384
South Africa	ZAF	0.235
Malaysia	MYS	0.052
Pakistan	PAK	0.049
Morocco	MAR	0.044
Philippines	PHL	0.043
Vietnam	VNM	0.042
Colombia	COL	0.042
Hungary	HUN	0.039
Kazakhstan	KAZ	0.036
Poland	POL	0.034

Table 17: Pre-Treatment Outcome Comparison (Log Production, 2010–2015).

Year	Taiwan (Actual)	Synthetic Taiwan	Gap
2010	12.623	12.622	0.001
2011	12.746	12.698	0.049
2012	12.734	12.733	0.001

Year	Taiwan (Actual)	Synthetic Taiwan	Gap
2013	12.733	12.817	-0.084
2014	12.846	12.818	0.028
2015	12.769	12.792	-0.024

Table 18: Placebo Inference: Post/Pre RMSPE Ratios (2016–2019 post-period).

Unit	Pre-Treatment RMSPE	Post-Treatment RMSPE	Post/Pre Ratio
Taiwan	0.042	0.270	6.37
Donor median (N=11)	0.193	0.507	3.00

Taiwan’s ratio of 6.37 exceeds 10 of 11 donor placebos, yielding a permutation p-value of approximately 0.17. While Taiwan’s gap is directionally consistent with the security channel, the permutation test does not achieve conventional significance thresholds, reflecting the small donor pool and COVID-confounded post period. We interpret the synthetic control as suggestive corroboration rather than confirmatory evidence.

6. Adjacent Comparisons and Future Research

Given the analysis above, two broader comparisons would strengthen the argument further, and we flag them as the most productive directions for future work.

6.1 The Cross-Sector Test

Within Taiwan, if the restriction on Chinese EVs were driven by industry protection rather than security, we would expect it to be concentrated in sectors where Taiwanese firms have domestic market share to defend. The observable pattern runs in the opposite direction: Taiwan has restricted Chinese goods across multiple sectors with strategic sensitivity (such as telecommunications equipment, drones, and digital infrastructure) where no meaningful domestic producer lobby exists. The

EV restriction fits this broader securitization pattern rather than a sector-specific lobbying outcome. A systematic cross-sector analysis — coding Taiwan’s import restrictions across two-digit HS categories by both domestic producer presence and strategic sensitivity — would allow a direct test of this prediction and substantially strengthen the causal argument. That analysis requires data collection beyond the scope of this paper but is a natural next step.

6.2 The South Korea Comparison

The South Korea comparison points to a second extension. South Korea faced the same Chinese economic coercion after the 2017 THAAD deployment and registered the same lesson about Beijing’s willingness to weaponize interdependence. Yet South Korea did not impose maximum restriction on Chinese EVs. The difference is structural: Hyundai and Kia export heavily into China and face retaliatory exposure if Seoul restricts Chinese imports. South Korea perceived the threat but faced a much higher cost of acting on it. This contrast confirms the paper’s interaction logic — the security channel predicts restriction when perceived threat is high and retaliatory export exposure is low — but a full paired process trace comparing Taiwan and South Korea’s domestic deliberations would sharpen this finding considerably. That comparison lies beyond the scope of a single cross-national study but represents the clearest path toward stronger causal identification of the security channel.

7. Conclusion

This paper has argued that national security is a necessary component of any explanation for cross-national variation in policy restrictiveness toward Chinese EV imports. GH has partial explanatory power, but the two anomalies identified in Section 1 — Japan’s openness despite the world’s second-largest auto sector, and Taiwan’s maximum restriction despite a minimal one — require the security channel to explain. Adding security alignment improves model fit in specifications that correctly address EU clustering, while the full-sample result is attenuated by the globally competitive au-

tomaker exception and the EU's single-policy clustering problem. Taiwan is the largest positive residual in the industry-only model — the observation that makes the security channel necessary.

The Taiwan process trace provides mechanism identification. Five pieces of evidence support the security channel: declining domestic capacity (eliminating the industry-lobbying explanation), rising cross-strait trade dependence creating weaponizable vulnerability, Beijing's demonstrated willingness to coerce (tourism restrictions after 2016), Taiwan's acceptance of consumer welfare costs (consistent with security-motivated restriction), and temporal sequencing linking security escalations to policy tightening. The synthetic control analysis provides quantitative corroboration: Taiwan's log production fell approximately 0.256 log units below the counterfactual trajectory in the 2016–2019 window, consistent with security-driven suppression of industrial integration.

The 2x2 framework — which interacts ICE industry size with EV adoption level to determine which theoretical channel is operative — represents the paper's theoretical contribution. It generates quadrant-specific predictions and identifies where each channel has explanatory power: Q2 for Grossman-Helpman, Q3 for the security premium, Q4 as the boundary condition where consumer welfare overwhelms both channels. Taiwan in Q3, with maximum restriction despite absent industry lobbying, is the sharpest test of H2.

The policy implications are significant. As Chinese EV manufacturers expand globally, every importing country will face the same trade-off: cheaper vehicles versus supply-chain dependence on a geopolitically assertive state. The Grossman-Helpman model predicts restriction only where large domestic industries resist. The security-augmented model predicts that the frontier of restriction will be determined by security alignment — and that countries perceiving China as a geopolitical threat will restrict more than their economic interests alone would predict. Taiwan is the limiting case of this logic today. It may not be the last.

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Declaration of AI Use

This project used AI tools in the following ways: Claude (Anthropic) assisted with R code debugging and the replication script, helped identify and format data sources in the appendix, and

provided grammar and clarity edits on completed prose. AI was not used to generate arguments, write any substantive portion of the paper, conduct the literature review, or produce statistical results. All research questions, theoretical frameworks, empirical specifications, and interpretations are the authors' own. The ordered logit models, synthetic control analysis, and process trace were designed and executed by the authors; AI assisted only in code formatting and output presentation.

Appendix A: Data Sources and Variable Construction

Variable	Source	Year	Notes
A.1 Production	OICA	2022 (pre-treat)	Universal shock_year = 2023
A.2 Sales	OICA	2022	For computing prod/sales ratio
A.5 Domestic EV	IEA + manual	2024	Binary: domestic EV brand exists
A.6 EV Adoption	IEA GEV 2025	Avg 2022–24	Share of new cars sold that are electric
A.8 GDP per capita	World Bank WDI	Most recent	Current USD
AVE Tariff	DV sheet + reg.	2025	Continuous: country-specific effective rate
B.1 China Trade	IMF DOTS	2024	Import/export share with China
B.3 BRI	Manual coding	2024	Binary membership status
C.1 Alliance	ATOP/SIPRI + manual	2024	Ordinal 0–5, US-China alignment
C.3 UNGA	BSV ideal points	2024	Alignment score; Taiwan = N/A
D.1 V-Dem LDI	V-Dem v16	2024	Liberal Democracy Index 0–1
D.3 Ideology	DPI 2024	2024	Ruling party L=1/C=2/R=3
DV Code	GTA + WTO + national	2025	Authors' 0–3 ordinal coding

Appendix B: Robustness Checks

B.1 Continuous-AVE Specification (OLS)

The continuous ad valorem equivalent (AVE) specification provides a complementary check on the ordered logit results. Taiwan is coded at 200 percent (a conservative prohibitive-tariff equivalent for an administrative ban). OLS with heteroskedasticity-robust standard errors yields $R^2=0.033$ (adj. $R^2=-0.011$), directionally consistent but with low explanatory power attributable partly to Brazil's extreme AVE coding (1,022%); the ordered logit is preferred for the main analysis.

Table 11: OLS Results: AVE Dependent Variable (Robust SEs, N=47)

Term	Estimate	Std. Error	t value	p value
Intercept	-79.210	104.453	-0.758	0.452
Log Production	12.733	10.648	1.196	0.238
Alliance Ordinal	-6.529	5.558	-1.175	0.246

B.2 DV Sensitivity Analysis

Collapsing DV=2 and DV=3 into a single "high restriction" category yields $\log_prod=0.162$ (SE=0.133, $t=1.22$) and $alliance_ord=0.038$ (SE=0.188, $t=0.20$), confirming that the NTB category boundary is not mechanically driving results. The directional pattern of the coefficients is unchanged.

B.3 Multicollinearity Diagnostics

All predictors fall below the conventional VIF threshold of 5 (maximum: V-Dem LDI = 4.22), ruling out multicollinearity as a concern. Figure 5 shows the alliance coefficient remains directionally consistent across all four sequential specifications (M1 through M4), further confirming that the full-sample null is not produced by collinear regressors.

B.4 UNGA-Inclusive Specification (Option B, Taiwan Excluded)

Because Taiwan lacks a UN ideal point score, the primary specifications use alliance ordinal only (Option A, Taiwan included). As a robustness check, we estimate Option B — UNGA-inclusive, Taiwan excluded (N=46). UNGA alignment shows a positive coefficient (+1.41, $t=1.53$) consistent with H2. Taiwan's out-of-sample predicted DV is 2 with $P(DV=3)=4.4\%$, convergent with the leave-Taiwan-out prediction under Option A (Table 9), confirming the anomaly is not an artifact of the alliance-only coding choice.

Appendix C: Replication Script

The script below reproduces all main results from `merged_analytical_v2.csv`. Run top-to-bottom in a clean R session. HTML comments flag code blocks whose output also appears as a formatted table in the main text; review before final submission to avoid duplication.

```
# =====  
# THE SECURITY PREMIUM -- REPLICATION SCRIPT  
# DPI-410, Harvard Kennedy School, Spring 2026  
# Reproducible from: merged_analytical_v2.csv  
# =====  
  
# --- 1. SETUP: PACKAGES & DATA -----  
  
pkgs <- c("tidyverse", "MASS", "scales", "knitr", "ggrepel", "patchwork",  
          "car", "sandwich", "lmtest", "kableExtra", "Synth")  
  
for (p in pkgs) if (!requireNamespace(p, quietly = TRUE)) install.packages(p)  
suppressPackageStartupMessages(lapply(pkgs, library, character.only = TRUE))  
  
df_raw <- read_csv("merged_analytical_v2.csv", show_col_types = FALSE)  
df <- df_raw |>  
  mutate(  
    dv          = as.integer(dv_code),  
    dv_factor   = ordered(dv),
```

```

log_prod    = log(as.numeric(A1_production) + 1),
production_k = as.numeric(production_k),
ev_adopt    = as.numeric(ev_adoption_pct),
log_gdp     = log(as.numeric(A8_gdp_pc) + 1),
alliance_ord = as.numeric(C1_alliance_ord),
unga        = as.numeric(C3_unga_alignment),
vdem        = as.numeric(D1_vdem_ldi),
ave         = as.numeric(ave_tariff)
) |>
filter(!is.na(dv), !is.na(log_prod), log_prod > 0,
       !is.na(alliance_ord), !is.na(vdem))

# Unified helper: tidy coefficient table from any polr model
extract_model <- function(model, name) {
  if (is.null(model))
    return(tibble(Model = name, Term = "FAILED",
                  Estimate = NA, SE = NA, z = NA, p = NA))
  ct <- coef(summary(model))
  rows <- !grepl("\\\\|", rownames(ct))
  tibble(Model = name,
         Term      = rownames(ct)[rows],
         Estimate  = ct[rows, "Value"],
         SE        = ct[rows, "Std. Error"],
         z         = ct[rows, "t value"],
         p         = 2 * pnorm(abs(ct[rows, "t value"]), lower.tail = FALSE))
}

# --- 2. ORDERED LOGIT M1--M7 (Option A: Alliance only, Taiwan included) -----
m1 <- polr(dv_factor ~ log_prod, data = df, Hess = TRUE)
m2 <- polr(dv_factor ~ alliance_ord, data = df, Hess = TRUE)
m3 <- polr(dv_factor ~ log_prod + alliance_ord, data = df, Hess = TRUE)
m4 <- tryCatch(
  polr(dv_factor ~ log_prod + alliance_ord + vdem + log_gdp,
       data = df, Hess = TRUE),
  error = function(e) NULL)
m5 <- tryCatch(

```

```

    polr(dv_factor ~ log_prod * alliance_ord, data = df, Hess = TRUE),
    error = function(e) NULL)
m6 <- tryCatch(
  polr(dv_factor ~ log_prod + alliance_ord + global_comp,
    data = df, Hess = TRUE),
  error = function(e) NULL)
m7 <- tryCatch(
  polr(dv_factor ~ log_prod + alliance_ord + vdem + log_gdp + global_comp,
    data = df, Hess = TRUE),
  error = function(e) NULL)

# Coefficient table (thresholds suppressed)
# POSSIBLE DUPLICATE: see Table 5 in main text -- review before submitting
bind_rows(
  extract_model(m1, "M1: Industry"),
  extract_model(m2, "M2: Security"),
  extract_model(m3, "M3: Combined"),
  extract_model(m4, "M4: Full"),
  extract_model(m5, "M5: Interaction"),
  extract_model(m6, "M6: + Global Comp"),
  extract_model(m7, "M7: Full + Global Comp")
)

# Model fit and LR test
# POSSIBLE DUPLICATE: see Table 6 in main text -- review before submitting
model_list <- Filter(Negate(is.null),
  list("M1"=m1, "M2"=m2, "M3"=m3, "M4"=m4, "M5"=m5, "M6"=m6, "M7"=m7))
tibble(
  Model = names(model_list),
  AIC = sapply(model_list, AIC),
  LogLik = sapply(model_list, function(m) as.numeric(logLik(m))),
  N = sapply(model_list, nobs)
)
anova(m1, m3) # LR test: does security improve over industry baseline?

# --- 3. OLS / AVE ROBUSTNESS -----

```

```

# POSSIBLE DUPLICATE: see Table 11 in Appendix B -- review before submitting

df_ols <- df |> filter(!is.na(ave))

m_ols3 <- lm(ave ~ log_prod + alliance_ord, data = df_ols)

ols_result <- coeftest(m_ols3, vcov = vcovHC(m_ols3, type = "HC3"))

p_vals <- as.numeric(ols_result[, "Pr(>|t|)"])

ols_tbl <- data.frame(
  Term      = rownames(ols_result),
  Estimate = round(as.numeric(ols_result[, "Estimate"]), 3),
  SE        = round(as.numeric(ols_result[, "Std. Error"]), 3),
  t         = round(as.numeric(ols_result[, "t value"]), 3),
  p         = round(p_vals, 3),
  sig       = case_when(p_vals < 0.01 ~ "***", p_vals < 0.05 ~ "**",
                        p_vals < 0.10 ~ "*",   TRUE      ~ ""),
  stringsAsFactors = FALSE
)

cat("R-squared:", round(summary(m_ols3)$r.squared, 3), "\n")

# --- 4. EU CLUSTERING ROBUSTNESS -----

# POSSIBLE DUPLICATE: see Table 8 in main text -- review before submitting

eu_isos <- c("AUT", "BEL", "BGR", "HRV", "CYP", "CZE", "DNK", "EST", "FIN", "FRA",
            "DEU", "GRC", "HUN", "IRL", "ITA", "LVA", "LTU", "LUX", "MLT", "NLD",
            "POL", "PRT", "ROU", "SVK", "SVN", "ESP", "SWE")

# (a) EU collapsed: all 27 members -> one production-weighted observation

df_eu_coll <- df |>
  filter(!(iso3 %in% eu_isos)) |>
  bind_rows(
    df |> filter(iso3 %in% eu_isos) |>
      summarise(iso3 = "EU27", country = "EU (aggregate)",
                dv = 2, dv_factor = ordered(2),
                log_prod = log(sum(exp(log_prod), na.rm = TRUE)),
                alliance_ord = mean(alliance_ord, na.rm = TRUE),
                vdem = mean(vdem, na.rm = TRUE),
                log_gdp = mean(log_gdp, na.rm = TRUE))
  )

m3_eu <- tryCatch(

```

```

  polr(dv_factor ~ log_prod + alliance_ord, data = df_eu_coll, Hess = TRUE),
  error = function(e) NULL)
extract_model(m3_eu, "M3 EU-collapsed")

# (b) EU excluded
m3_no_eu <- tryCatch(
  polr(dv_factor ~ log_prod + alliance_ord,
    data = df |> filter(!(iso3 %in% eu_isos)), Hess = TRUE),
  error = function(e) NULL)
extract_model(m3_no_eu, "M3 EU-excluded")

# --- 5. LEAVE-TAIWAN-OUT CROSS-VALIDATION -----
# POSSIBLE DUPLICATE: see Table 9 in main text -- review before submitting
m3_no_twn <- polr(dv_factor ~ log_prod + alliance_ord,
  data = df |> filter(iso3 != "TWN"), Hess = TRUE)
twn_row <- df |> filter(iso3 == "TWN")
predict(m3_no_twn, newdata = twn_row, type = "class") # observed = 3
predict(m3_no_twn, newdata = twn_row, type = "probs")

# --- 6. DV SENSITIVITY + VIF -----
# DV collapsed (0/1/2+)
df_3cat <- df |> mutate(dv_3 = ordered(pmin(dv, 2)))
m3_3cat <- tryCatch(
  polr(dv_3 ~ log_prod + alliance_ord, data = df_3cat, Hess = TRUE),
  error = function(e) NULL)
extract_model(m3_3cat, "M3 3-category DV")

# VIF (from M4 full specification)
if (!is.null(m4)) {
  m4_lm <- lm(dv ~ log_prod + alliance_ord + vdem + log_gdp, data = df)
  vif(m4_lm)
}

# Option B: UNGA-inclusive, Taiwan excluded
df_B <- df |> filter(!is.na(unga))
m_optB <- tryCatch(

```

```

polr(dv_factor ~ log_prod + alliance_ord + unga, data = df_B, Hess = TRUE),
error = function(e) NULL)

extract_model(m_optB, "Option B: UNGA")

if (!is.null(m_optB)) {
  twn_imputed <- tibble(log_prod = df |> filter(iso3=="TWN") |> pull(log_prod),
                        alliance_ord = 4,
                        unga = mean(df_B$unga[df_B$alliance_ord >= 3], na.rm=TRUE))

  predict(m_optB, newdata = twn_imputed, type = "probs")
}

# --- 7. SYNTHETIC CONTROL (Abadie, Diamond & Hainmueller 2010) -----
years <- 2010:2024

prod_panel <- tribble(
  ~iso3, ~country,
  "TWN", "Taiwan",
  "MYS", "Malaysia",
  "PAK", "Pakistan",
  "PHL", "Philippines",
  "ZAF", "South Africa",
  "MAR", "Morocco",
  "ARG", "Argentina",
  "COL", "Colombia",
  "POL", "Poland",
  "HUN", "Hungary",
  "VNM", "Vietnam",
  "KAZ", "Kazakhstan"
) |>

mutate(production = list(

  c(303456, 343296, 339038, 338720, 379223, 351085, 309531, 291563, 253241, 251304, 245615, 265320, 261263, 285962, 275156),

  c(567715, 533695, 572150, 601407, 596600, 614671, 513445, 460140, 564800, 571632, 485186, 481651, 702225, 774600, 790347),

  c(152970, 162194, 165700, 134145, 148746, 229686, 214650, 250800, 269700, 186751, 117375, 238702, 235454, 79513, 128449),

  c(65625, 53921, 55360, 52260, 106938, 112493, 116686, 141251, 79763, 95094, 67297, 85874, 92223, 110350, 126571),

```

```

c(472049,532545,539424,545913,566083,615658,599004,589951,610854,631983,447218,499087,555889,633337,599755),

c(42066,59477,108743,167452,231986,288329,345106,376286,402085,394652,248430,403007,464864,535825,559645),

c(716540,828771,764495,791007,617329,533683,472776,472158,466649,314787,257187,434753,536893,610725,506571),
c(109333,41714,32466,70686,71137,78070,79036,74994,75476,78020,47281,40764,51455,34700,23778),

c(609474,708133,654756,690159,693904,760603,781837,789729,809646,498158,451382,439421,483840,612882,555346),

c(211461,202800,217840,222400,437599,495370,472000,505400,430988,498158,406497,394302,441729,507225,437045),

c(42286,31181,40470,37576,48871,50000,236161,236161,237000,250000,165568,163250,232410,177435,175661),
c(3176,8195,19186,40684,40162,14477,10651,19071,31545,49400,74831,92417,112540,146989,144624)
)) |>
unnest_longer(production) |>
mutate(year = rep(years, n_distinct(iso3)),
       log_production = log(production + 1)) |>
left_join(df_raw |> select(iso3, gdp_pc = A8_gdp_pc) |>
         mutate(gdp_pc = as.numeric(gdp_pc)), by = "iso3") |>
mutate(gdp_pc = if_else(is.na(gdp_pc), mean(gdp_pc, na.rm = TRUE), gdp_pc))

unit_lookup <- prod_panel |> distinct(iso3) |> mutate(unit_id = row_number())
prod_panel <- prod_panel |> left_join(unit_lookup, by = "iso3")
tw_n_id <- unit_lookup |> filter(iso3 == "TWN") |> pull(unit_id)
donor_ids <- unit_lookup |> filter(iso3 != "TWN") |> pull(unit_id)

sc_result <- tryCatch({
  dp <- dataprep(
    foo = as.data.frame(prod_panel),
    predictors = "gdp_pc",
    predictors.op = "mean",
    special.predictors = list(
      list("log_production", 2010, "mean"),
      list("log_production", 2012, "mean"),
      list("log_production", 2014, "mean")
    )
  )
})

```

```

),
dependent          = "log_production",
unit.variable      = "unit_id",
time.variable      = "year",
treatment.identifier = twn_id,
controls.identifier = donor_ids,
time.predictors.prior = 2010:2015,
time.optimize.ssr   = 2010:2015,
time.plot          = 2010:2024
)

capture.output(so <- synth(dp, quadopt = "ipop"))
synth_y <- as.numeric(dp$Y0plot %*% so$solution.w)
actual_y <- as.numeric(dp$Y1plot)
path_df <- tibble(year = as.integer(rownames(dp$Y1plot)),
                  actual = actual_y, synthetic = synth_y)
list(dp = dp, so = so, path_df = path_df, success = TRUE)
}, error = function(e) list(success = FALSE, msg = e$message))

# Predictor balance (Table A), donor weights (Table B), pre-treatment fit (Table C)
# -- see hardcoded tables in main text (Tables 15-17) --

```